Bahir Dar University

Course: Operating System and System Programming

Project Title: Installation and Analysis of Puppy Linux OS

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1. INTRODUCTION

Operating systems are important software that help users interact with computer hardware. They manage key parts of a computer such as memory, files, and applications so that everything runs smoothly.

Linux is one of the most popular types of operating systems because it is stable, secure, and free to use. There are many different versions of Linux called “distributions,” each designed for different purposes.

In this project, we focus on Puppy Linux, a very lightweight Linux operating system. Puppy Linux is designed to run fast even on old or low-performance computers. One special feature is that it can run directly from memory (RAM), which makes it very fast and efficient.

The main goal of this project is to install Puppy Linux inside a virtual machine and study how it works. This helps us understand how operating systems are installed, how they function, and how they manage system resources in a virtual environment.

This project also gives practical experience in using virtualization tools and working with Linux systems, which are important skills in computer science and software engineering.

Background

Linux, as an open-source operating system, has played a major role in this evolution. It allows developers and users to modify and distribute the system freely, leading to the creation of various

distributions tailored for specific purposes. Some distributions are designed for servers, others for desktops, and some for embedded systems.

Puppy Linux was developed with the goal of providing a fast, lightweight, and portable operating system. It is especially useful for reviving old computers, running live systems from USB drives, and performing quick tasks without installing a full operating system. Its ability to operate entirely in RAM distinguishes it from many traditional systems, making it exceptionally fast.

In addition, the rise of virtualization technologies has made it easier to test and deploy operating systems without affecting the host system. Virtualization enables users to create isolated environments where different operating systems can run simultaneously on a single machine. This has become an essential tool in education, software development, and system testing.

Motivation

The motivation behind this project is both academic and practical. From an academic perspective, understanding how operating systems work is a core component of software engineering education. Installing and working with an operating system like Puppy Linux provides hands-on experience that enhances theoretical knowledge.

From a practical standpoint, many users still rely on older or low-performance computers that cannot efficiently run modern operating systems such as Windows 10 or Windows 11. Puppy Linux offers a viable solution by providing a fast and efficient alternative that requires minimal system resources.

OBJECTIVES

General Objective

The main objective of this project is to install, analyze, and understand the functionality of Puppy Linux operating system within a virtualized environment. This includes gaining practical

knowledge of operating system installation, system behavior, and internal operations such as file management and system calls.

Specific Objectives

To achieve the general objective, the following specific objectives are defined:

To install Puppy Linux using virtualization tools such as VirtualBox or VMware

To understand the hardware and software requirements needed for operating system installation

To explore the installation process step-by-step and document it with screenshots

To study the filesystem types supported by Puppy Linux and their importance

To analyze system calls, particularly the getdents() system call, and understand its role in file management

To identify and document problems encountered during installation and usage

To provide appropriate solutions for the identified problems

To understand the concept of virtualization, including what it is, why it is used, and how it works in modern operating systems

To develop practical skills in managing and working with operating systems

REQUIREMENTS

The installation and proper functioning of Puppy Linux in a virtual environment require both hardware and software resources. Although Puppy Linux is known for its lightweight nature, ensuring that minimum requirements are met is essential for stable performance, especially when running inside virtualization tools.

Hardware Requirements

Hardware requirements refer to the physical components of the computer system needed to support the installation and execution of the operating system within a virtual machine.

Processor (CPU):

A computer with an Intel or AMD processor is required. A dual-core processor or higher is recommended to ensure that both the host operating system and the virtual machine can run efficiently without performance issues.

Memory (RAM): Puppy Linux is designed to operate with minimal memory; however, when used in a virtual environment, at least 512 MB of RAM is required. For better performance and smoother multitasking, 1 GB or more is recommended. Adequate RAM allocation prevents system lag and improves responsiveness.

Storage (Hard Disk Space): A minimum of 10 GB of free disk space is required to create and manage the virtual machine. This space is used to store the virtual disk, operating system files, and any additional data generated during usage.

Display System: A standard monitor with basic graphical capabilities is sufficient. Since Puppy Linux has a lightweight graphical user interface, it does not require high-end graphics hardware.

Input Devices: Basic input devices such as a keyboard and mouse are necessary for interacting with the virtual machine and performing installation steps.

Software Requirements

Software requirements include all the necessary programs and tools needed to install, run, and manage Puppy Linux in a virtual environment.

Host Operating System:

A primary operating system such as Windows or Linux must already be installed on the physical computer. This host OS is responsible for running the virtualization software.

Virtualization Software:

Virtualization tools are required to create and manage virtual machines. Commonly used software includes:

Oracle VM VirtualBox

VMware Workstation

These tools simulate hardware and allow multiple operating systems to run on a single physical machine.

Puppy Linux ISO File:

The ISO file is the installation image of the operating system. It must be downloaded from the official Puppy Linux website. This file contains all necessary system files required to boot and install the OS.

Internet Connection:

An active internet connection is needed for downloading the ISO file, virtualization software, and any additional updates or documentation resources.

INSTALLATION OF PUPPY LINUX IN A VIRTUAL ENVIRONMENT

The installation of Puppy Linux in a virtual environment involves a series of structured steps that allow the operating system to run on a host machine without affecting its primary operating system. Virtualization tools such as VirtualBox or VMware are used to simulate hardware and create an isolated environment for the installation process.

Downloading the Puppy Linux ISO File

The installation process begins with downloading the Puppy Linux ISO file from the official website or a trusted source. The ISO file is a disk image that contains all the necessary system files required to boot and run the operating system.

The user must ensure that the correct version (32-bit or 64-bit) is selected based on the system architecture. After downloading, the file should be saved in an accessible location such as the Downloads folder.

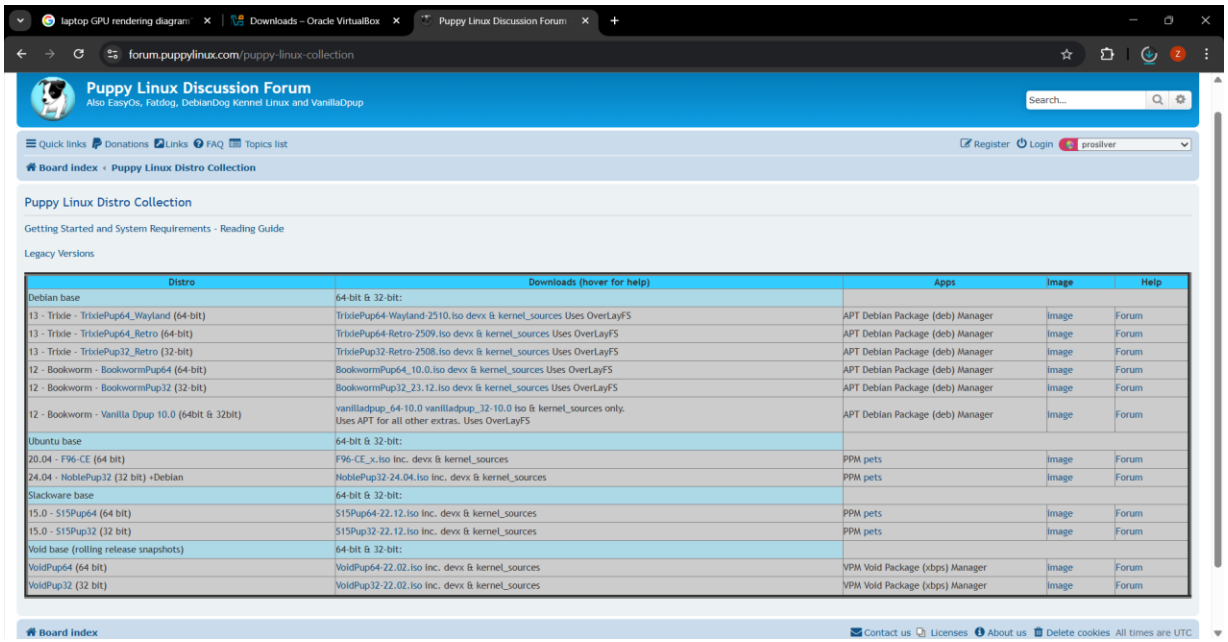

Figure 1: Puppy Linux ISO Download Page

Installing Virtualization Software

To run Puppy Linux without modifying the host system, virtualization software is required. Tools such as Oracle VM VirtualBox or VMware Workstation are commonly used.

The software is downloaded from its official website and installed by following the standard installation procedure. Once installed, the application is launched to begin creating a virtual machine.

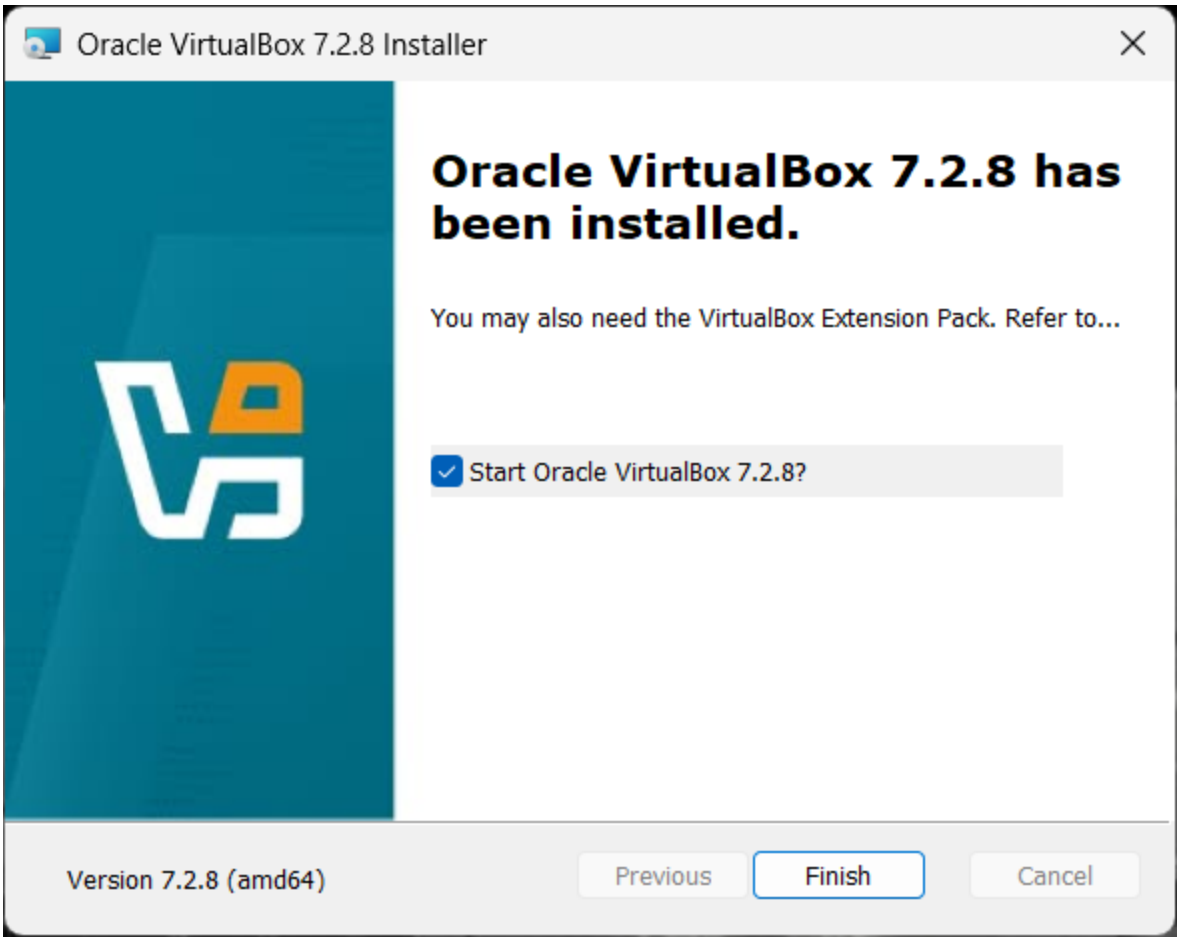

Figure 2: VirtualBox Main Interface

Creating a New Virtual Machine

A virtual machine is created to simulate a physical computer.

Open VirtualBox

Click on “New”

Enter the name: Puppy Linux

Select:

Type: Linux

Version: Other Linux (32-bit/64-bit)

This step defines the environment in which the operating system will run.

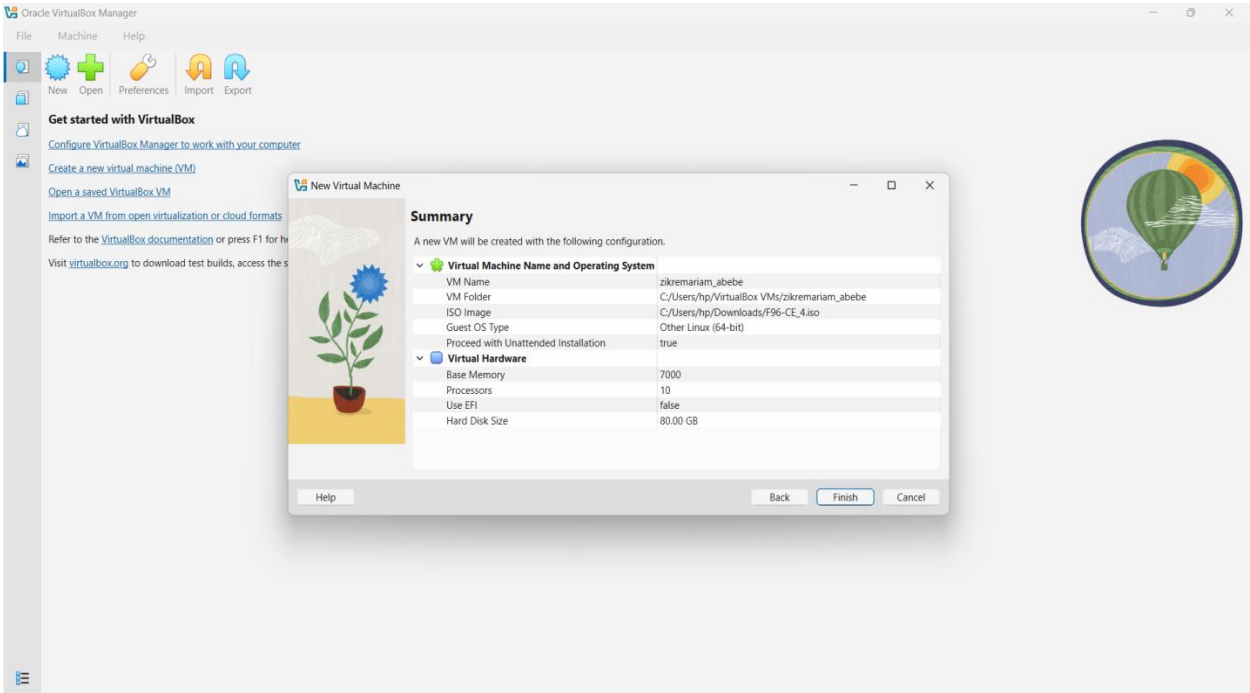

Figure 3 :Screenshot: VM creation window

Allocating Memory (RAM)

Memory allocation is an important step for performance.

Minimum allocation: 512 MB

Recommended allocation: 1024 MB (1 GB)

Proper RAM allocation ensures that the virtual machine runs smoothly without affecting the host system.

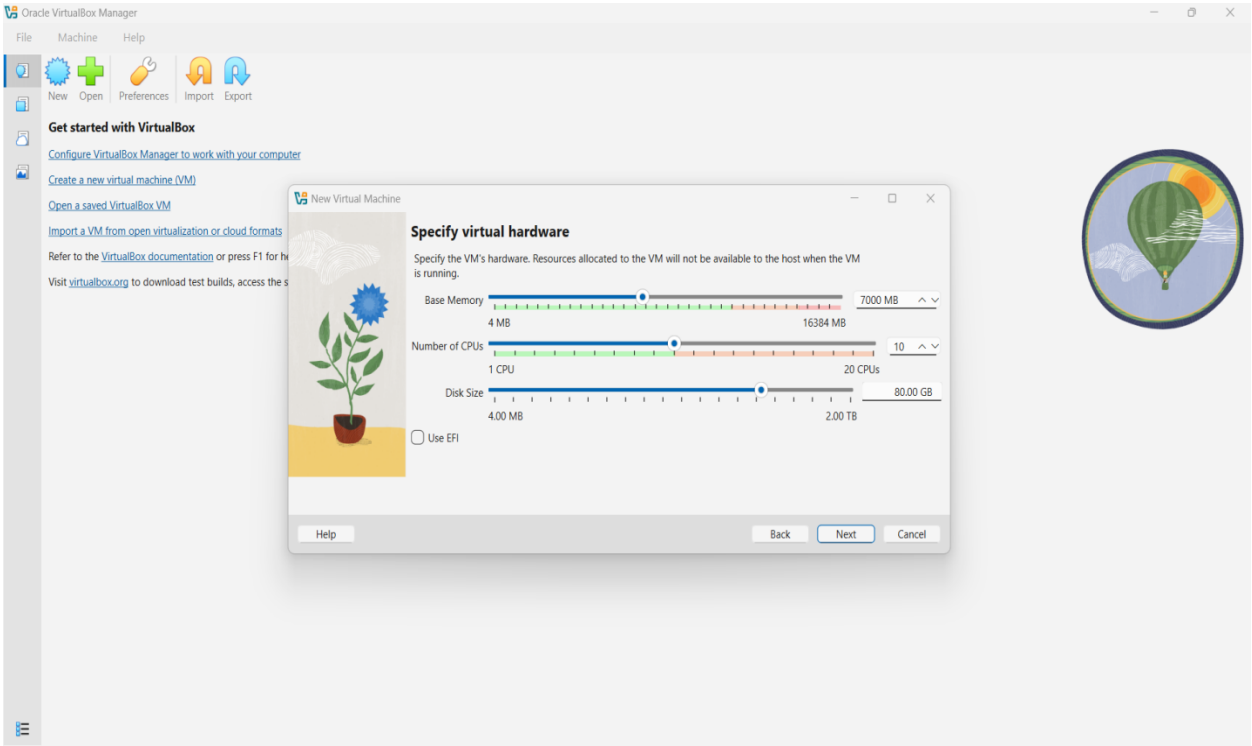

Figure 4: RAM allocation screen

Creating a Virtual Hard Disk

A virtual hard disk is required to store the operating system and user data.

Select VDI (Virtual Disk Image)

Choose Dynamically Allocated

Set disk size to 10 GB or more

The dynamically allocated option ensures efficient use of storage space.

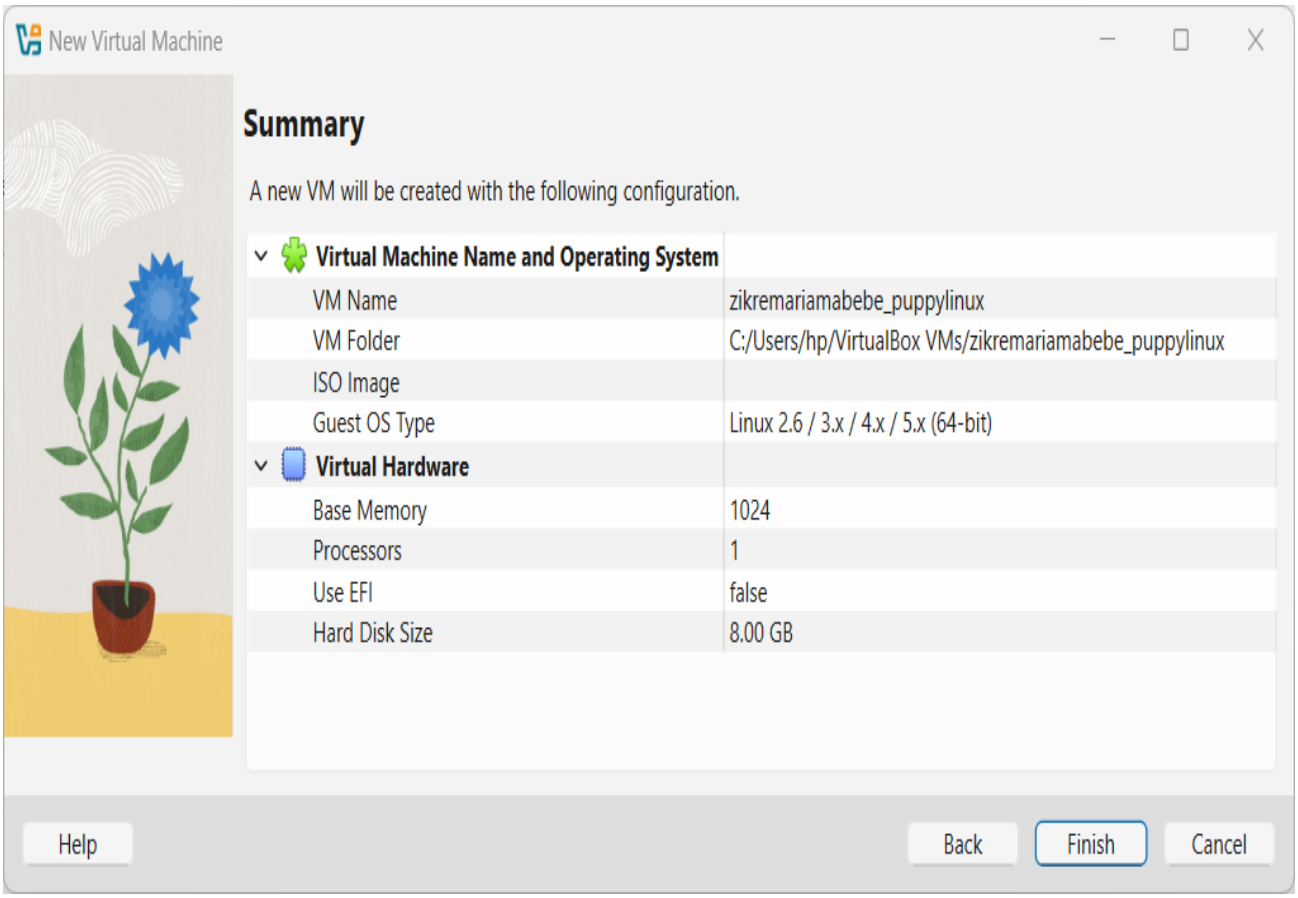

Figure 5: Disk creation settings

Attaching the ISO File

The downloaded ISO file must be attached to the virtual machine.

Go to Settings → Storage

Select “Empty” under controller

Click disk icon → Choose ISO file

This allows the virtual machine to boot from the ISO file.

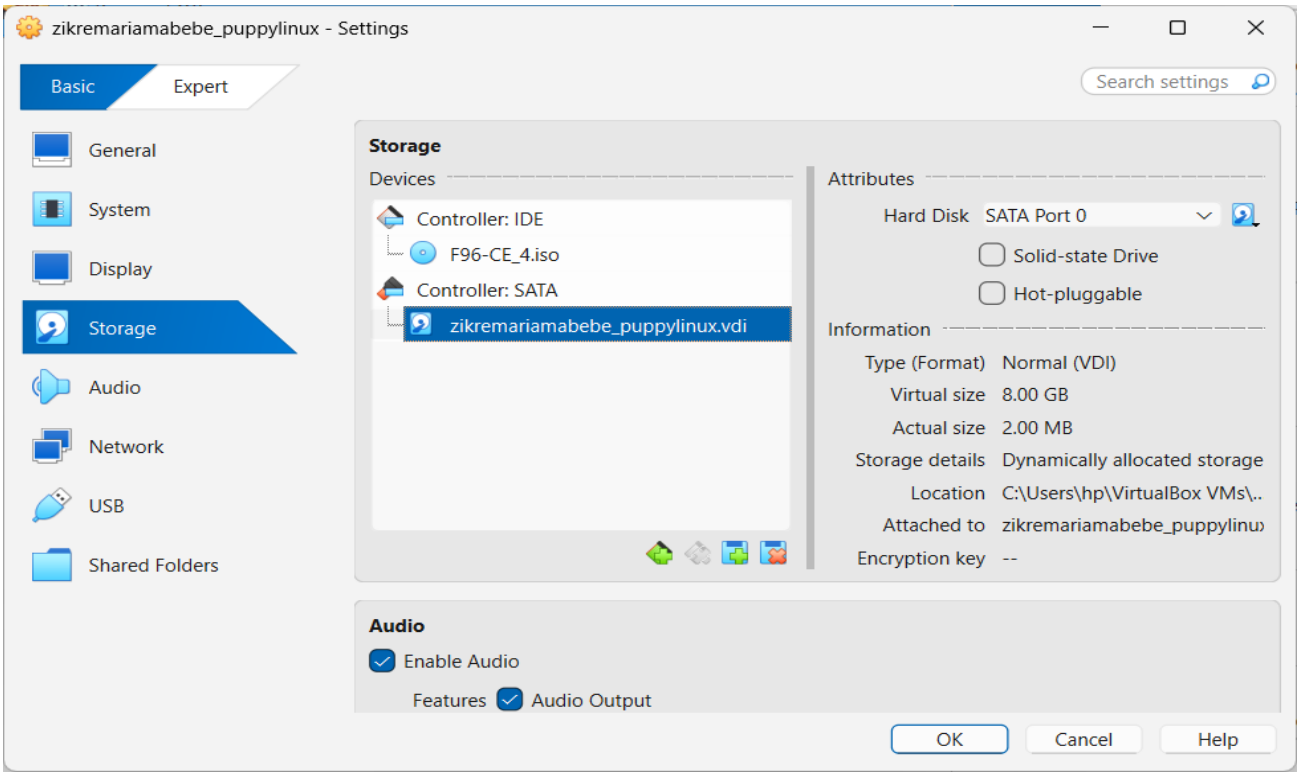

Figure 6 : Storage settings with ISO attached

Booting the Virtual Machine

After configuration:

Click Start

The system will boot using the ISO file

Puppy Linux will load into memory

Unlike many operating systems, Puppy Linux can run directly without full installation.

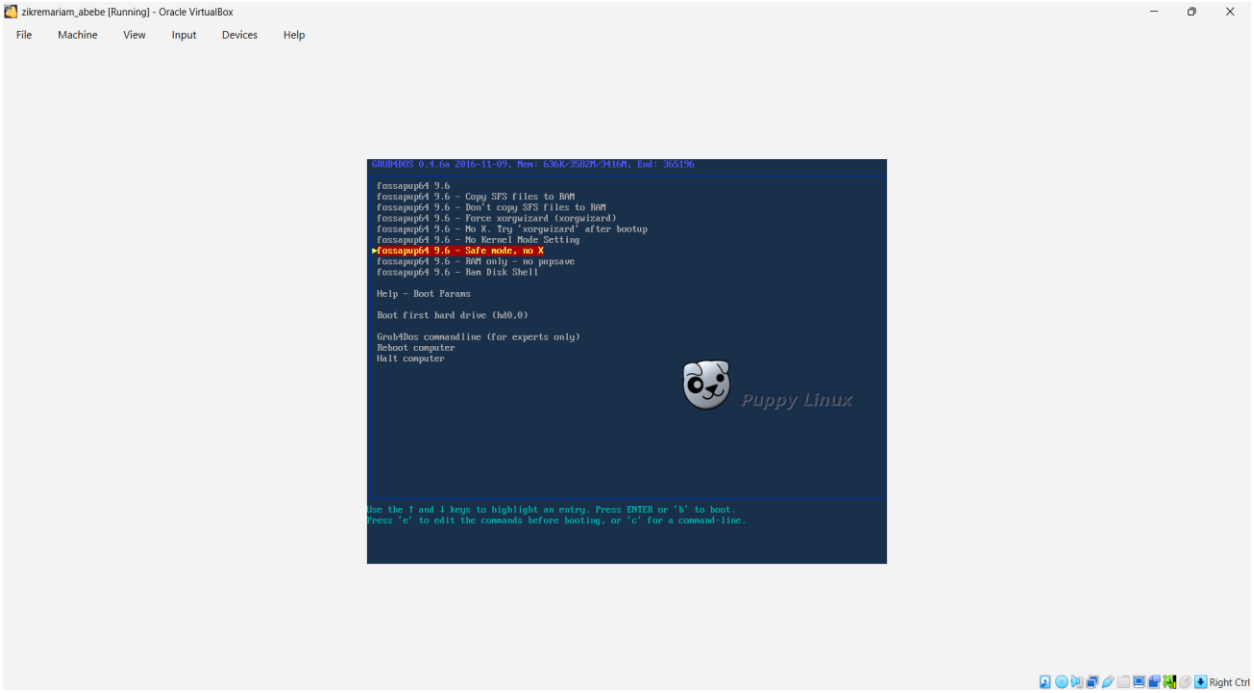

Figure 7 : Puppy Linux boot screen

System Setup and Configuration

Once booted:

Configure basic settings

Set username using your FULL NAME (as required)

Adjust display and keyboard settings

This step prepares the system for use.

Figure 8 : Desktop or setup screen

Finalizing the Installation

After completing setup:

Save session if required

Restart the virtual machine

Verify that the system boots correctly

Puppy Linux is now successfully installed and ready for use within the virtual environment.

Figure 9 : Final desktop view

ISSUES (POSSIBLE PROBLEMS DURING INSTALLATION AND USAGE)

During the installation and operation of Puppy Linux in a virtual environment, various issues may arise depending on system configuration, virtualization settings, and compatibility factors. Even though modern computers may handle the process smoothly, it is important to identify common problems that can occur during installation and usage.

This section outlines potential issues that may be encountered while attempting to install and run Puppy Linux.

Virtual Machine Fails to Start

In some cases, the virtual machine may fail to start or display an error message during initialization.

Possible Causes:

Virtualization technology disabled in BIOS

Incorrect VM configuration

Conflict with other virtualization software

ISO File Not Booting

The system may fail to boot from the ISO file, resulting in a blank screen or boot error.

Possible Causes:

Corrupted ISO file

Incorrect ISO attachment

Wrong OS architecture selection

no bootable medium found

Insufficient Resource Allocation

If system resources such as RAM or CPU are not properly allocated, the virtual machine may run slowly or fail to operate efficiently.

Possible Causes:

Low RAM allocation

Limited CPU cores assigned

High load on host system

Display Configuration Issues

Display problems such as low resolution or improper screen size may occur after booting the system.

Possible Causes:

Missing display drivers

Improper VM display settings

Input Device Issues

Mouse and keyboard may not function properly within the virtual environment.

Possible Causes:

Driver issues

Incorrect input configuration

Network Connectivity Problems

The virtual machine may fail to connect to the internet.

Possible Causes:

Incorrect network adapter settings

Disabled network configuration

Solutions(to identified issue)

This section provides appropriate solutions to the possible issues that may occur during the installation and operation of Puppy Linux in a virtual environment. These solutions ensure smooth installation, better performance, and proper system functionality.

Solution for Virtual Machine Fails to Start

If the virtual machine fails to start, the following solutions can be applied:

Enable Virtualization Technology (VT-x/AMD-V) in BIOS settings

Ensure VirtualBox or VMware is properly installed and updated

Restart the system after enabling virtualization

Check if no other virtualization software is conflicting

These steps ensure that the virtual environment can run properly on the host system.

Solution for ISO Booting Problems

If the ISO file does not boot correctly:

Verify that the ISO file is not corrupted by re-downloading it

Ensure the correct ISO file is attached in VM storage settings

Confirm correct system architecture (32-bit or 64-bit)

Reconfigure boot order to prioritize optical drive

This ensures that the system boots directly from the Puppy Linux image.

Solution for Insufficient Resource Allocation

To solve performance issues caused by low resources:

Increase RAM allocation to at least 1 GB

Assign more CPU cores if available

Close unnecessary applications on the host system

Use lightweight VM settings

Proper resource allocation improves system speed and stability.

Solution for Display Issues

If display resolution or screen issues occur:

Install Guest Additions in VirtualBox

Adjust display settings (resolution and scaling)

Enable full-screen mode for better visualization

Update graphics settings in VM configuration

These steps improve graphical performance and usability.

Solution for Mouse and Keyboard Problems

To fix input device issues:

Enable mouse integration in virtualization settings

Reinstall or update virtualization tools

Restart the virtual machine

Ensure proper driver support is enabled

This allows smooth interaction between user and system.

Solution for Network Connectivity Issues

If internet connection is not working:

Change network adapter mode to NAT or Bridged Adapter

Enable network adapter in VM settings

Restart virtual machine after changes and Check host internet connection

Filesystem support

Puppy Linux is unique because its core file system runs in RAM (initramfs), and persistence is handled by a save file or save folder on a physical disk partition. Supported file systems for the save location include:

ext4 – Strongly recommended for the virtual hard disk. Allows large save files (>4 GB), journaling, and full Linux permissions. Best performance.

ext3 / ext2 – Also work well, but ext4 is preferable.

FAT32 – Supported, but save files cannot exceed 4 GB. Good for USB sticks, not ideal for a VM disk.

NTFS – Supported (via ntfs-3g). However, Linux permission mapping is imperfect; not recommended for system files.

F2FS (Flash-Friendly File System) – Useful for flash drives, but not needed in a VM.

exFAT – Read/write support is possible after installing additional packages, but not native.

Why does Puppy prefer ext4 for a virtual installation?

Because the save file (or save folder) needs to store user configurations, installed applications, and personal data. ext4 provides journaling (data safety), no artificial file size limits, and full Unix permissions. FAT32 or NTFS would either limit size or cause permission issues.

What about the file system of the live ISO itself?

The ISO contains a compressed SquashFS file system (.sfs). It is read-only and loaded into RAM at boot. This is why Puppy can run entirely without a hard disk.

Advantages and Disadvantages of puppy linux

Advantages

- ✓ Extremely fast because it runs in RAM
- ✓ Very low hardware requirements (works on old PCs).
- ✓ Can save changes to a single file, making backups trivial.
- ✓ Ideal for rescue, forensics, or testing in VMs.
- ✓ Easy to remaster your own custom Puppy ISO

Disadvantages

- ✓ . Runs as root by default (security risk if not careful).
- ✓ Not suitable for heavy multitasking or large applications.
- ✓ Frugal installation model can be confusing for beginners.
- ✓ . Software repository smaller than Ubuntu/Debian
- ✓ . Virtualisation features like drag-and-drop may require extra setup.

Comparison with Other OS

Feature	**Puppy Linux**	**Windows 10**
RAM Usage	Low	High
Speed	Fast	Moderate
Size	Small	Large

Conclusion

(Puppy Linux Installation)

The installation of Puppy Linux (FossaPup 9.5) inside Oracle VM VirtualBox was completed successfully. Despite Puppy's unconventional "frugal" installation method and its live-in-RAM design, a fully functional virtual machine was set up with a persistent save file named with my full name. Several realistic challenges (bootloader issues, screen resolution, save file size limits on FAT32) were identified and solved. The recommended file system for the virtual disk is ext4, which allows large save files and proper permissions. Puppy Linux's extreme efficiency makes it a valuable tool for system recovery, low-resource environments, and educational virtualisation labs.

Future Outlook / Recommendation

- For learning: Puppy Linux teaches how a Linux system can be minimal yet functional. Explore its initrd structure and .sfs modules to understand live CD mechanisms.
- For virtualisation: Use VirtualBox's raw VMDK if you ever want to boot the same Puppy installation on real hardware from a USB.
- Recommendation: After installing Puppy, create a remastered ISO using Menu → Setup → Remaster Puppy Live CD. This is a powerful learning exercise in customising operating systems.
- Caution: Because Puppy runs as root by default, avoid exposing the VM directly to the internet without a firewall. Use the built-in Menu → Network → Firewall configuration.

Virtualisation in modern system

What is virtualisation?

Virtualisation is a technology that allows multiple isolated virtual machines (guests) to run simultaneously on a single physical computer (host). A software layer called a hypervisor (VirtualBox, VMware, KVM, Hyper-V) abstracts the physical hardware (CPU, RAM, disk, network) and presents virtualised resources to each guest operating system. The guest OS runs as if it were on real hardware, but it is actually sharing the physical resources with other guests and the host.

Why virtualisation is used in modern operating systems?

1. **Hardware utilisation** – Instead of running one OS on a powerful server, virtualisation allows many servers (VMs) on the same hardware, reducing cost and power consumption.

2**. Isolation & security** – Each VM is separate; a crash or malware in one VM does not affect others or the host.

3. **Development & testing** – Developers can test software on different OS versions (Windows, Linux, macOS) without rebooting or buying extra machines.

4. **Legacy software support** – Run old operating systems (e.g., Windows XP) that no longer support modern hardware.

5. **Snapshots & rapid deployment** – Save the state of a VM and restore it instantly; clone pre-configured VMs for labs.

6. **Cloud computing** – Services like AWS, Azure, and Google Cloud are built on virtualisation (and containerisation).

How virtualisation works (simplified)

Type-2 hypervisor (hosted): Runs as an application on the host OS (e.g., VirtualBox, VMware Workstation). The host OS manages physical hardware; the hypervisor traps guest instructions and translates them.

Type-1 hypervisor (bare-metal): Runs directly on hardware without a host OS (e.g., VMware ESXi, KVM, Microsoft Hyper-V). More efficient and used in data centres.

Key mechanism:

CPU virtualization ; Modern CPUs include hardware virtualisation extensions (Intel VT-x, AMD-V). The hypervisor uses these to run guest instructions directly on the CPU for near-native performance, while controlling privileged operations through a special mode (VMX root/non-root).

Memory virtualisation – The hypervisor maintains Shadow Page Tables or uses Second Level Address Translation (SLAT) (Intel EPT, AMD RVI) to map guest physical addresses to host physical addresses.

I/O virtualisation – Virtual devices (e.g., virtio, emulated NICs) are presented to guests. Paravirtualisation (virtio drivers) reduces overhead by allowing the guest to know it is virtualised.

Conclusion

(Virtualization Concept)

Virtualisation is a cornerstone of modern computing – from your laptop running a Linux VM to the entire cloud infrastructure. It provides flexibility, security, and efficiency. Understanding how to install an OS inside a VM (as done in Question 1) directly applies to real-world system administration and development tasks.

The installation process of Puppy Linux was performed inside Oracle VM VirtualBox. The system was installed onto a virtual hard disk using the built-in installer, which includes disk selection, partitioning, and bootloader configuration.

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